

Implicit theories of courage

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Abstract

What is courage? This question garners significant interest and attention but little empirical research. An operational definition of courage is essential to good research, yet no consensus definition has fully emerged. This article systematically investigates people's conceptions of courage and courageous behavior through a series of studies employing well-grounded implicit methodologies. The organizational structure and components of courage are investigated using a response-generating task (Study 1) and alternate methods (Studies 2 and 3), followed by an experimental approach to determine if people actually use their implicit theories in their evaluations of others (Study 4). Collectively, these studies reveal an organizational structure of people's implicit theories of courage. Further, they indicate that people apply their implicit theories accurately in evaluating others.

Keywords: *Courage, courageous behavior, implicit theories, prototypes*

Then, Laches, suppose that we first set about determining the nature of courage... Tell me, if you can, what is courage

Socrates (Plato, trans. 1987)

Introduction

In Denver, Colorado, a 9-year-old boy disregarded his mother's desperate plea to run as two masked men attempted to break into their family home. Filled with terror, yet displaying selfless disregard, he threw himself over his 4-year-old brother to shield him from the invaders. The boy was shot in the back, but his younger brother was not hit ("Denver Boy Honored," 2005). Halfway across the country in Cedar Grove, Wisconsin, a 44-year-old man tried to save his 6-year-old daughter who had fallen through the thin layer of ice covering a skating pond. During his attempt, he fell in. Both father and daughter tragically died ("Wisconsin Father, Daughter Die," 2005). As we ponder these true-life events, we might ask ourselves: were these acts courageous? If so, why? And if not, why not?

What *is* courage? As one's perceptions of the events above show, courage is at least, in part, in the eye of the beholder. What one person might view as

courageous (jumping into an ice-covered lake) another might view as foolhardy. In some instances, such as spying, what one group views as courageous another might view as treacherous. And in cases of suicide bombers, what appears as courageous to one group might be viewed as evil to other groups. More to our point, then, is the question of what people *think* courage is. The current investigation is an attempt to bring us toward an understanding of courage, its nature, and its use.

Definitions of courage

Scholars, politicians, and laypeople have all entered the debate over how to define courage. Some believe that American culture over the last 30 years or so has defined courage. That is, courage has been "attributed to all manner of actions that may indeed be admirable but hardly compare to the conscious self-sacrifice on behalf of something greater than self-interest" (McCain & Salter, 2004, p. 13). At the same time, others would argue that courage can be attributed not only to the occasional and isolated act of rescue or self-sacrifice, but also to everyday acts (Evans & White, 1981; Putman, 2001; Woodard, 2004). Defining courage in terms of character

strengths (bravery, persistence, integrity, and vitality) of human goodness and excellence, the positive psychology movement endeavors to view courage through the lenses of positive subjective experiences, positive individual traits, and institutions that enable positive experiences and positive traits (Peterson & Seligman, 2004; Seligman & Csikszentmihalyi, 2000).

In the course of investigating the concept of courage, we uncovered numerous definitions or descriptions of courage of which 29 are listed in Table I. Hemingway's description of courage as "grace under pressure" (cited in Lopez, O'Byrne, & Peterson, 2003, p. 191) marks the simplest, yet eloquent, example. A more complex, multidimensional definition of courage as "the disposition to voluntarily act, perhaps fearfully, in a dangerous circumstance, where the relevant risks are reasonably appraised, in an effort to obtain or preserve some perceived good for oneself or others recognizing that the desired perceived good may not be realized" (Shelp, 1984, p. 354) anchors the other extreme. Though numerous definitions of courage provide a rich foundation from which to build, we remain at a loss for an operational definition of this construct on which to base sound explicit theories. Even with all of the attempts to define courage, we have not advanced the domain to an agreed-upon conceptual definition. Lopez and colleagues correctly assert that "though we have been able to parse out the different types of courage by establishing between-brand differences, we have been less successful at determining the elements or components of courage. Thus, what is common to all brands remains unclear" (Lopez et al., 2003, p. 189). Operational definitions of constructs of interest are essential to good research and a lack of coherence in courage definitions is an obstacle to the advancement of research. Therefore, the search for a common structure of courage (apparently underlying the disagreement as to people's implicit theories of courage) forms a major focus of the current investigation.

Theories of courage

Why are there so many definitions of courage, and why is courage so haphazardly defined? This should come as no surprise in the context of implicit-theory studies which demonstrate a diffuse understanding of implicit theories. Theories of courage can be divided into explicit- and implicit-theoretical categories, and both are important to the advancement of the psychological field. One might consider, however, that explicit theories of courage may simply be the implicit theories of courage researchers (Ardelt, 2005).

Explicit theories of courage

Explicit theories are constructions of psychologists that are based on data collected from people performing tasks presumed to measure the construct under investigation. In an explicit approach to the measurement of courage, Cox, Hallam, O'Conner and Rachman (1983) measured decorated (identified as courageous) bomb operators' physiological responses to fear and stress compared to non-decorated (noncourageous) operators' responses. They found distinctive physiological responses under stress for decorated and nondecorated bomb operators, indicating that past courageous behavior in a particular situation will reduce one's physiological responses to fear and stress in similar situations. Replication experiments sampling bomb-disposal operators (O'Connor, Hallman, & Rachman, 1985) and follow-up experiments studying British veteran paratroopers (Macmillan & Rachman, 1987) supported these initial findings.

Implicit theories of courage

According to Sternberg (1985) and Sternberg, Conway, Ketron and Bernstein (1981), implicit theories are people's own cognitive constructions. Such theories reside in people's heads, and need to be discovered rather than invented because they already exist. The goal of research on implicit theories is to find out the form and content of people's informal theories. The data of interest are people's communications (in whatever form) regarding their notions as to the nature of the construct. Implicit theories are important in psychology and in the world because most judgments people make of each other are based on their implicit theories (Sternberg, 1987). For example, there is no formal test of courage; people evaluate each other's courage on the basis of their implicit theories of what courage means. Even constructs which can be measured using explicit theories, such as in the assessment of intelligence, the overwhelming majority of judgments are based on implicit theories, such as when people evaluate others' intelligence on the basis of job interviews, colloquium presentations, published articles, interactions on dates, and so forth.

The study of naturalistic concepts has furthered understanding of several psychological constructs, such as love and commitment (Barnes & Sternberg, 1997; Fehr, 1988), wisdom (Holliday & Chandler, 1986; Sternberg, 1985), creativity (Sternberg, 1985), intelligence (Sternberg, 1985; Sternberg et al., 1981), and moral exemplars (Walker & Hennig, 2004), and is now beginning to further our understanding of courage (Evans & White, 1981; O'Byrne, Lopez, & Peterson, 2000; Woodard, 2004).

Table I. Selected definitions and descriptions of courage.

Source	Definitions and descriptions
<i>American Heritage Dictionary</i>	The state or quality of mind or spirit than enables one to face danger with self-possession, confidence, and resolution; bravery (1985).
Aquinas*	Defined <i>fortitude</i> as firmness in mind in enduring or repulsing whatever makes steadfastness outstandingly difficult, that is, particularly serious dangers, primarily sustaining action to overcome fears of bodily harm and death and secondarily in persevering in attacking.
Aristotle*	Defined <i>andreia</i> (military courage) as the disposition to act appropriately in situations that involve fear and confidence: rationally determined mean between cowardice and foolhardiness.
Cavanagh and Moberg	Courage, also called fortitude or bravery, is the ability to endure what is necessary to achieve a good end, even in the face of great obstacles (1999, p. 2).
Clancy	Courage is likely defined as a willingness to face tough choices as well as overcoming the fear associated with them (2003, p. 132).
Evans and White	An empirical definition of courage probably involves three important attributional dimensions: (a) the fear level of the person making the attribution; (b) the perceived fear level of the attributee; and (c) salient features of the situation e.g., objective risk involved and so on (1981, p. 420).
Finfgeld*	Being courageous involves being fully aware of and accepting the threat of a long-term health concern, solving problems using discernment, and developing enhanced sensitivities to behavior consists of taking responsibility and being productive.
Gergen and Gergen*	To be courageous, then, is to remain steadfast within the bosom of those relationships from which one's sense of personal esteem and identity are derived.
Gould	Courage is revealed in three dimensions: (1) fear; (2) appropriate action; and (3) a higher purpose.
Haitch*	Courage is two-sided: there is an aspect of standing firm or fighting, and an aspect of accepting intractable realities . . . courage is the psychic strength that enables the self to face danger and death.
Hemingway*	Grace under pressure.
Hobbes*	The contempt of wounds and violent death. It inclines men to private revenges, and sometimes to endeavor the unsettling of public peace.
Kant*	Defined <i>fortitudo</i> as the capacity and the resolved purpose to resist a strong but unjust opponent; and with regard to the opponent of the moral disposition within us.
Kennedy*	(Describing senators with political courage) men whose abiding loyalty to their nation triumphed over personal and political considerations.
Kilmann, O'Hara and Strauss	A courageous act in an organization includes five essential properties: (1) member has free choice to act; (2) member experiences significant risk; (3) member assess the risk as reasonable; (4) member's contemplated act pursues excellence or other worthy aims and (5) member proceeds despite fear with mindful action (2005).
Klein and Napier	Courage involves five factors: candor (speak and hear the truth), purpose (pursue lofty and audacious goals), rigor (invent disciplines and make them stick), risk (empower, trust, and invest in relationships), and will (inspire optimism, spirit, and promise) (2003).
Kohut*	Oppose the pressures exerted on them and remain faithful to their ideals and themselves.
McCain and Salter	Defined courage as an act that risks life or limb or other very serious personal injuries for the sake of others or to uphold a virtue: a standard often upheld by battlefield heroics but one that is certainly not limited to martial valor (2004, p. 14).
Mencius (Mengzi)	Distinguished between types of courage, seeing some as "petty," those concerned exclusively with personal honor; and "great," those grounded in and oriented toward the good. "Those who know that they are in the right are justified in their cause and this provides them with the motivation to confront and engage even the greatest of dangers" (cited in Ivanhoe, 2002, p. 68).
O'Byrne, Lopez and Peterson*	Dispositional psychological courage is the cognitive process of defining risk, identifying and considering alternative actions, and choosing to act in spite of potential negative consequences in an effort to obtain "good" for self or others recognizing that this perceived good may not be realized.
Plato*	The ability to remember what is worth prizing and what is worth fearing.
Putman*	Facing the fears associated with the loss of psychological stability.
Rachman	Willing and able to approach a fearful situation despite the presence of subjective fear and psychophysiological disturbances (1990, p. 12).
Seligman*	The capacity to rise to the occasion.
Shelp*	The disposition to voluntarily act, perhaps fearfully, in a dangerous circumstance, where the relevant risks are reasonable appraised, in an effort to obtain or preserve some perceived good for oneself or others recognizing that the desired perceived good may not be realized.
Shepela et al.	Courageous resistance: selfless behavior in which there is high risk/cost to the actor, and possibly the actor's family and associates, where the behavior must be sustained over time, is most often deliberative, and often where the actor is responding to a moral call (1999, p. 789).
Snyder*	Extraordinary behavior in ordinary times.
Walton	Courage consists of three characteristics: (1) careful presence of mind and deliberate action, (2) difficult, dangerous, and painful circumstances, and (3) a morally worthy intention . . . at the agent's personal risk and suffering (1986, p. 3).
Woodard	Courage is defined as the ability to act for a meaningful (noble, good, or practical) cause, despite experiencing the fear associated with perceived threat exceeding the available resources (2004, p. 174).

Note: The definitions attributed to sources annotated by a star (*) are from Lopez, O'Byrne and Peterson (2003). Copyright 2003 by the American Psychological Association. Adapted with permission of the author.

Implicit-theory studies of courage. In a response-generation study examining how the average person views courage, O'Byrne et al. (2000) asked 97 people the question, "What is courage?" They found that laypeople's views of courage varied considerably. Participants described courage as "taking action," "standing up for what one believes in," "sacrificing," and "facing threats/fears/challenges and overcoming obstacles." Philips (2004) asked 65 people the same question, "What is courage?" discovering similar themes regarding people's notions about courage. Courage requires sacrifice, risk, and overcoming fear for a good purpose. To have courage is to endure, persevere, and overcome. These studies, not unlike the present article's Study 1, provide an empirical rationale for the numerous definitions of courage. The responses provide only foundation from which we can ascertain the components by which people evaluate courage, and the organization underlying individuals' conceptions of courageous behavior. If left as stand-alone studies the results would indicate the potential for many subjective definitions of courage, however, the underlying structure of courage would remain undetermined.

In efforts to develop a measurement scale of courage, Woodard (2004) also utilized implicit-theories methodologies. Ten experts with varied areas of specialty in the field of psychology generated statements considered representative of assessing the construct of courage, defined as "the ability to act for a meaningful (noble, good, or practical) cause, despite experiencing the fear associated with perceived threat exceeding the available resources" (Woodard, 2004, p. 174). Two hundred participants then rated their agreement with the statements on a 5-point scale. Each statement was also accompanied by a fear-rating question to establish the level of fear that the respondent might associate with the situation depicted in the statement, and by a third question asking the respondents whether they had experienced the situation posed in the statement. Factor analysis revealed four dimensions labeled: (a) endurance for positive outcomes—including behaviors such as "acts despite bullying as a minority"; (b) dealing with groups—including behaviors such as "help grieving family" and "rejection by others for goals"; (c) acting alone—including behaviors such as "accept job despite criticism" and "avoid confronting my own pain"; and (d) physical pain/breaking social norms—including behaviors such as "intervene in domestic dispute" and "endure pain for political secrets." The factors were found to have a strong correlation with each other, suggesting a relationship among the ideas that contributed to deciding whether to act in a courageous manner. The concepts

represented by the factors have further illuminated the basic manner in which people conceive of the construct of courage.

In a series of interview-based studies in the nursing domain, chronically ill adolescents (Haase, 1987), middle-aged adults (Finfgeld, 1998), and older adults (Finfgeld, 1995) were asked to describe a situation in which they thought they were courageous. They were instructed to describe their thoughts, feelings, and perceptions as they remembered experiencing them. These findings point to the development of attitudes and coping methods, highlighting the process of becoming and being courageous in the face of chronic illness.

Additional studies of courage have also implemented implicit-theories methodologies. Evans and White (1981) focused on the role of fear in people's implicit theories of bravery (used synonymously with courage). They concluded that courage probably involved, in part, not only the perceived fear of the target actor, but also the fear level of the person making the evaluation. Szagun (1992) and Szagun and Schauble (1997) used structured interviews to investigate juvenile, adolescent, and adult conceptualizations of courage. In their developmental approach, they revealed that courage was viewed as less physical and increasingly non-physical with age. Themes of deliberate or intentional cognitive processes, overcoming or persisting despite fear, risk, and moral acts were indicated.

Caveat to the study of courage

As a complex, multi-dimensional construct, courage may be better understood as an exceptional response to specific external conditions or circumstances than as an attribute, disposition, or character trait (generally acknowledged as long-term and stable across situations) which appears to be the intent of many definitions and descriptions of courage. Perhaps researchers are on firmer ground when they label someone as "courageous" based on their actions in a given situation. In other words, given a specific situation, researchers can say that one acted or responded "courageously," if appropriate.

Conversely, many notable researchers (see O'Byrne et al., 2000; Peterson, 2006; Peterson & Seligman, 2006; Shelp, 1984) through explicit statement, definition, or description of courage favor a more dispositional courage. For example, Peterson and Seligman (2004) through their classification of virtues and character strengths have provided an extraordinary foundation for looking at the trait-like qualities of courage. These researchers have quite nicely described courage as a virtue achieved through the character strengths of bravery,

persistence, integrity and vitality. Peterson and Seligman (2004) do note, however, that their conceptualization of bravery (what many courage researchers and philosophers label “courage”) is difficult to measure because of the tonic or phasic nature of different “types” of courage. Their Values in Action Inventory of Strengths (VIA-IS) “asks about typical responses rather than those situated in the appropriate settings for the display of given strengths. Further, no one can reasonably nominate friends or acquaintances as paragons of given strengths unless they have happened to observe them in the appropriate settings” (Peterson & Seligman, 2004, p. 633). This appears to indicate that even if researchers hold the view of dispositional courage, the situation still matters.

Therefore, while not excluding the possibility of a courageous disposition or a relationship with character, there does seem to be limited transferability between a “courageous act” and a “courageous actor.” We should not necessarily conclude that individuals have a courageous disposition based on their actions in one situation. We can not appropriately infer that the same individual will act courageously in a different situation. “In short, to say that someone is a ‘courageous person,’ when the speaker means to generalize to long-term dispositions or character habits, may be very shaky conjecture indeed on the basis of a single action” (Walton, 1986, p. 221).

Summary

Why are we studying implicit theories of courage? Sternberg et al. (1981) remind us that implicit theories tell us about individuals’ views of what a construct such as courage is. Wegner and Vallacher echo this point, stating that “the layman . . . lives by his theory The systems we call ‘implicit theories’ are the individual’s reality” (Wegner & Vallacher, 1977, p. 21). Generally, implicit theories are theories of word usage and the lay understanding of the construct behind the word. In this case, courage is of extreme interest to a vast number and variety of people, from academic scholars, to military members, and to the ordinary citizen. It is important to know people’s notions of courage’s nature and if it is both possible and beneficial to cultivate courage. Implicit theories are one approach toward developing a comprehensive scheme for understanding and describing positive subjective experiences, positive individual traits, and positive institutions (Peterson, 2006; Peterson & Seligman, 2004). In sum, our motivation was to systematically investigate people’s conceptions of courage and courageous behavior

through a series of studies employing well-grounded implicit methodologies (Bluck & Glück, 2005).

Although similar studies have used response-generation (O’Byrne et al., 2000; Philips, 2004) and item-rating tasks (Woodard, 2004) to investigate the construct of courage, our studies greatly expand upon this pioneering work. In Study 1 and Study 2, we employed independent, non-overlapping samples from two very distinct graduate and undergraduate populations (i.e., Yale University and the United States Air Force Academy [USAFA]) to ascertain the nature and use of courage. These samples were obtained from the Yale University population at large and through two classes required by all USAFA students. Furthermore, Study 3 extended beyond what others have accomplished by introducing alternative methods to confirm an underlying structure of courage. To our knowledge, Study 4 is the first study to investigate the application of courage and its components in the evaluation of others. The essential first step is to determine an underlying structure of courage. Does this structure hold, however, when people actually use these components in their evaluation of others’ levels of courage? Study 4 attempted to address this question. In sum, we investigated the organizational structure and components of courage using a response-generation task (Study 1) and alternate methods (Study 2 and 3) followed by an experimental approach to determine if people actually use their implicit theories in their evaluations of others (Study 4).

This sequence of studies provided an understanding of people’s implicit theories of the psychological construct of courage in everyday terms, while serving as a launching point to further investigate the implicit organization of courageous behavior. Core behavioral components were revealed through these studies. As we uncover and understand this particular psychological construct, and understand implicit-theory convergence with more recent explicit and scholarly definitions, such as courage as the emotional character strengths (bravery, persistence, integrity, and vitality) that involve “the exercise of will to accomplish goals in the face of opposition, either external or internal” (Peterson, 2006, p. 143; Peterson & Seligman, 2004, p. 199; Snyder & Lopez, 2007, p. 19), we can begin to look at the potential benefits of creating programs to address the development of courageous behavior in individuals. Knowledge of courage and potential developmental programs could have a significant impact on the ordinary citizen, business professionals, and individuals in the military, in fire-fighting, and in law enforcement, as courageous behavior could be and often is called upon in our daily lives and in the line of professional duty.

Study 1

Study 1 was conducted to compile a master list of behaviors people regard as descriptive of an ideally courageous individual. A free-listing task was accomplished by a total of 175 undergraduate and graduate students (109 male and 56 female) from the USAFA and Yale University. Participants included 107 undergraduate students (79 men and 28 women) at the USAFA. Their ages ranged from 18 to 26 years of age ($M=20.0$ years, $SD=1.7$). In addition, 68 undergraduate and graduate students at Yale University (30 men and 38 women) participated in this study. The Yale contingent ranged from 18 to 38 years of age ($M=22.5$ years, $SD=3.7$). We recruited both undergraduate and graduate students at Yale to ensure adequate sample sizes.

Participants were asked to spend 5 to 10 minutes completing a free-listing task in which they were instructed to imagine observing behaviors of an ideally courageous person. They were then asked to list as many behaviors as they could think of that are associated with this ideal individual. It was emphasized that we are interested in behaviors, not attributes. We want to know what courageous people do (observable actions that make one courageous), not how they are or what they are like (inferred attributes/traits). Participants were provided with the following example: "Saying that a courageous person 'is unflappable' would not really describe behavior; however, saying this person 'remains calm in pressure situations' would describe behavior." A raw-form master list of 1118 behaviors was compiled from the participants' responses. As a result of the substantial size of the master list, judgment rules were implemented to aid in distillation of the behaviors. Taking a conservative approach to avoid experimenter bias, we retained behaviors listed even once, although obvious redundancies were eliminated. (We also anticipated additional item reduction in Study 2.) The following judgment rules adapted from Walker and Hennig (2004) were implemented: (a) compound phrases were split if each behavioral descriptor could stand alone; (b) modifiers were dropped; (c) attributes were converted to behaviors, if possible; and (d) nonsensical, irrelevant, or inappropriate responses were eliminated (e.g., wears Right Guard®, wrestles a bear). The final list consisted of 639 items that participants deemed illustrative of an ideally courageous individual's behaviors. The responses indicated a substantial number of listed behaviors, which included willful actions involving risk, sacrifice, doing the right thing, and acting despite fear. The final list appears to provide a foundation from which we will begin to ascertain the structures, factors, or dimensions by which people evaluate

courage, and the organization underlying individuals' conceptions of courageous behavior itself in subsequent studies.

Study 2

In Study 1 we were able to obtain a large number of behaviors associated with courageous behavior. In Study 2, we asked participants to rate the degree to which these behaviors were associated with courage. This allowed us to integrate the myriad behaviors into a set of empirically driven factors that could be used in subsequent studies. Study 2 also allowed us to omit behaviors that did not appear to be generally associated with people's core perceptions of courage.

Method

Participants. A total of 126 undergraduate and graduate students participated in this second study. Participants included 59 undergraduate students, 41 men and 18 women, at the USAFA. Their ages ranged from 18 to 24 years of age ($M=20.75$, $SD=1.15$). In addition, a sample of 67 undergraduate and graduate students from Yale University participated. This sample included 19 men and 48 women. Their ages ranged from 18 to 56 years of age ($M=22.99$, $SD=5.10$). Participants received either US\$10 compensation or course credit for participation in this study.

Materials and procedure. Participants received an informed consent document and completed a background questionnaire eliciting responses to gender, age, and educational status. Participants were asked to spend 30 to 45 minutes completing the background questionnaire and rating task. Prior to the rating task, we instructed participants to imagine an ideally courageous person (i.e., someone who has just performed a courageous act) forming their own impressions of the ideal. We then asked them to rate how "distinctively characteristic" the 639 behaviors on the master list are in describing their ideally courageous person. All behaviors required ratings of 1 (low) to 9 (high).

Results

Missing data were considered missing at random and were therefore entered by group mean imputation. No single item was missing more than 3% of the data. Furthermore, no participant failed to rate more than 5% of the listed behaviors.

Basic statistics and prototypicality ratings. Given that a mix of undergraduate ($n=36$) and graduate ($n=31$) students at Yale University participated in this study in order to attain an appropriate sample size, a t -test was performed to see if this sample could be collapsed for future analyses. No significant differences were found for the mean distinctively characteristic ratings between undergraduate ($M=6.06$, $SD=1.15$) and graduate ($M=6.08$, $SD=1.21$) students. In addition, the correlation between Yale undergraduates and graduates across the distinctively characteristic ratings was 0.91. Thus, both the level and pattern of the responses were largely the same.

Overall, the mean rating of the items was 6.25 ($SD=1.16$). Ratings were also approximately equal between USAFA ($M=6.46$, $SD=1.13$) and Yale ($M=6.07$, $SD=1.17$) participants and between male ($M=6.18$, $SD=1.18$) and female ($M=6.32$, $SD=1.16$) participants. A 2 (academic institution) $\times$ 2 (gender) ANOVA was conducted. Both main effects and the interaction failed to reach significance. Ratings were extremely reliable, mid- to high-90s, both from the standpoint of Spearman–Brown split halves of participants (subject reliability) and of Spearman–Brown split halves of items (item reliability). The correlations between academic institutions and between genders across the distinctively characteristic ratings indicated strong associations, $r=0.91$ and $r=0.95$, $p<0.01$, respectively. Thus, the rating levels and pattern of ratings were similar for the sample populations.

As a result of the cumbersome nature of the original 639 items as input into a final analysis and of our recognized small sample size in comparison, only distinctively characteristic items with mean ratings of 7.0 or higher on the 1–9 distinctively characteristic scale were retained for further analysis. The number of items was reduced to a more manageable 131 items. Sternberg et al. cautioned that not all “characteristic” behaviors are important behaviors; for example, “eating is highly characteristic of intelligent people, but few people would see eating as central to defining their conceptions of such people” (Sternberg et al., 1981, pp. 43, 44). In the present study, however, laypeople deemed these particular behavioral items as “distinctive” or unique to courage (i.e., more central, core, or important) based on their ratings, rather than generally characteristic behaviors of an individual, courageous or not.

Structure and content of people’s conceptions of courage. Although the participants-to-variables ratio was noted as statistically problematic, exploratory factor analyses were performed using

principal-components extraction followed by varimax rotation (with Kaiser normalization) of the factorial axes. Separate analyses were first conducted for Yale and USAFA samples. Although there were differences between the exact factor structures obtained by the academic institutions, the factor solutions were indeed quite similar, mirroring the level and pattern ratings of the two samples, indicating remarkable similarities in their perceptions of courage. Based on this observation a combined factor analysis was then performed. Three factors were extracted (on the basis of scree analysis, strength, and interpretability) accounting for 23%, 19%, and 12% of the variance in the data, respectively, for a total of 54%. Table II shows the results of the factor analysis of the participants’ distinctively characteristic ratings of an ideally courageous person’s behavior. The three factors were labeled: (I) *self-focused response to affect and external circumstances*; (II) *non-physical/social-oriented acts—internal noble motivation*; and (III) *selfless sacrifice/risk—external noble motivation*. Behaviors with loadings above 0.63 (indicating 40% overlapping variance between the behavioral items and factor) were chosen for interpretability. Although any cut-off is subjective, Comrey and Lee (1992) suggested that interpretability of items loading in excess of 0.63 is very good. The highest loading items on each factor are listed in Table II.

The first factor, self-focused response to affect and external circumstances, included behaviors such as “endures tough situations,” “perseveres in face of obstacles,” “does not give in to fear when making decisions,” and “acts even in the presence of fear.” The second factor, non-physical/social-oriented acts—internal noble motivation, included behaviors such as “stands up to unjust social practices because of what one thinks is right,” “maintains honesty no matter others’ opinions,” “demonstrates integrity,” and “speaks up for one’s beliefs even under controversy.” The third factor, selfless sacrifice/risk—external noble motivation, included behaviors such as “sacrifices one’s life for another person’s safety,” “risks life to protect others,” and “risks one’s life for a greater good.”

Discussion

Analysis of data suggests that, for the most part, the listed behavioral items were viewed as quite characteristic of courageous individuals. No significant differences between mean ratings by academic institutions or genders were found. Further, the ratings were extremely reliable across both participants and items. All subject reliabilities exceeded 0.92 and all item reliabilities exceeded 0.97.

Table II. Factors underlying people's implicit conceptions of courage: "distinctively characteristic" ratings of an ideally courageous person.

Factor	Factor loading
(I) <i>Self-focused response to affect and external circumstances</i>	
Endures tough situations	0.83
Can handle rough situations	0.82
Handles tough, high-pressure situations	0.80
Perseveres in face of obstacles	0.80
Remains focused in high-stress environments	0.78
Remains composed in dangerous situations	0.78
Does not give in to fear when making decisions	0.77
Remains steadfast even when task requires unpleasant actions	0.77
Perseveres under pressure	0.76
Overcomes obstacles	0.76
Does not shy away when the situation demands action	0.75
Rises above pressure	0.74
Overcomes fears to perform dangerous tasks	0.74
Fights through pain and suffering	0.74
Acts even in presence of fear	0.74
Overcomes challenges	0.74
Refuses to bow to pressure	0.73
Does not freeze in pressured situations	0.73
Does not quit when "the going gets tough"	0.72
Faces that which causes fear	0.71
Remains calm when afraid	0.69
Pushes through stress	0.68
Does not back down when scared	0.68
Does not give up	0.67
(II) <i>Non-physical/social-oriented acts—internal motivation</i>	
Stands up to unjust social practices because of what one thinks is right	0.76
Stands up for the "right thing" against opposition	0.73
Maintains honesty no matter others' opinions	0.72
Acts upon one's values even if the values are not the norm	0.71
Demonstrates integrity	0.70
Does what is right even if no one will know	0.70
Stands up for what is right even if the majority disagrees	0.70
Does not fear to follow up on one's ideals	0.70
Acts based on moral principles rather than self-interest	0.70
Does what is right even if it is unpopular	0.70
Speaks up for one's beliefs, even under controversy	0.68
Does the right thing	0.68
Stands up for others' rights without thought of reward	0.68
Lives life according to personal ideals	0.68
Stands up for the rights of others	0.67
Stands up for what one believes is right	0.67
(III) <i>Selfless sacrifice/risk—external motivation</i>	
Risks life to protect others	0.86
Sacrifices self to save another	0.85
Risks one's own life to save others in need	0.82
Sacrifices one's life for another person's safety	0.80
Sacrifices self for good of someone else	0.75
Sacrifices one's life for another person's survival	0.74
Sacrifices health for another person's survival	0.74
Risks death to do something one knows is necessary	0.71
Risks their life for a greater good	0.67

Note: Proposed factor labels are italicized. The three factors account for 53.8% of the total variance in the data.

This high degree of internal consistency across participants (subject reliability) tells us there was considerable agreement as to what constituted courageous behavior. The high degree of internal consistency across items (item reliability) indicated that we avoided a hodgepodge of unrelated items; in other words, the items constituted a coherent set of behaviors.

The results of our factor analysis are intriguing. The analysis provided an underlying structure for how people judge others in terms of courageous behavior. These factors seem to point to general categories such as *external circumstances* (e.g., risk), *affective* (e.g., fear), and *motivational* (e.g., noble/worthy purpose) components. Furthermore, this analysis also provides compelling evidence of core components of courage against which definitions of courage may be evaluated. As implicit theories are about word usage, we can use the emergent structure to help determine the adequacy of previous definitions. For example, Shelp's (1984) description of courage seems to fully encapsulate the three emergent factors in this analysis. Other definitions are perhaps less fully adequate or complete in comparison. For instance, Rachman's (1990) definition of a courageous person as one "willing and able to approach a fearful situation despite the presence of subjective fear and psychophysiological disturbance" (p. 12) corresponds to just one emergent factor, self-focused response to affect and external circumstances; whereas Putman's (1997) definition of courage as "facing fears associated with the loss of psychological stability" also corresponds only to the self-focused response to affect and external circumstances factor, while introducing a "psychological stability" component undefined through the implicit theories of our population.

To conclude, this study shows that people do seem to have prototypes corresponding to the concept of courage that are organized into sensible factors. Based on this analysis, we would suggest that definitions of courage should reflect, in some form, the components of *risk*, *fear*, and *noble/worthy purpose*. This study also provides evidence for an underlying structure of courage against which definitions of courage may be evaluated. It also strengthens the foundation upon which further research may be based to reveal more about people's implicit theories of courage and how they use them.

Study 3

In Study 3, we asked participants to sort prototypical behaviors of courage based on their similarity in an attempt to reaffirm the organizational structure underlying the individual's conception of courageous behavior. Using a card-sorting task, analysed with

classical multidimensional scaling (CMDS) and cluster analysis, we confirmed the implicit and theoretical underpinnings of courage. Again, the components of risk, fear, and noble/worthy purpose emerged.

Method

Participants. In this second study, 65 participants completed a web-based “card-sorting” task. Tullis and Wood (2004) found that card sorts using a sample size of 30 correlated at 0.95 to a full-participant study ($N = 168$), with diminishing returns as the sample size increased (e.g., a 0.98 correlation for $n = 60$ with $N = 168$). Based on these observations, we recruited two samples of about 30 participants each. Thirty-five undergraduate students (26 men and 9 women) at the USAFA participated. In addition, participants included 30 undergraduate and graduate students (8 men and 22 women) from Yale University. Both undergraduates and graduates from Yale were recruited for this study to ensure an adequate sample size. The combined ages ranged from 17–29 years ($M = 20.98$, $SD = 2.23$). Participants received either course credit or US\$5 compensation for their participation.

Materials. Participants received an informed consent document and completed a background questionnaire eliciting responses to gender, age, and educational status. Next, the participants accessed WebSort[®] to complete a web-based, computerized similarity-sorting task in which they sorted 60 descriptive behaviors into as many or as few groups as they deemed appropriate. These behaviors, listed in Table III, were generated in the Study 1 free-listing task, and were identified from Study 2 as the top 60 prototypic behaviors of courage based on their mean distinctively characteristic ratings.

Procedure. Participants spent 20 to 30 minutes completing the background questionnaire and sorting task. All participants received the same set of 60 prototypic behaviors, and they were instructed to sort these behaviors into as many or as few categories as they deemed appropriate on the basis of which behaviors were “likely to be found together” in a person. The participants were not told in advance what the behaviors had in common (i.e., that they are descriptive behaviors of courage). All responses were electronically recorded.

Table III. Prototypicality (characteristicness) ratings of the behavioral items for an ideally courageous person.

Behavioral descriptors	<i>M</i>
Risks one's own safety to save others in need	8.15
Does the right thing even if it is hard to do	7.97
Fights for what one believes in	7.89
Does what is right even if no one will know	7.82
Acts even in presence of fear	7.75
Risks one's own life to save others in need	7.71
Stands up for what one believes in the face of strong opposition	7.71
Stands strong in the face of adversity	7.69
Sacrifices one's life for another person's safety	7.63
Does the right thing regardless of consequences to self	7.60
Stands up for what is right even if the majority disagrees	7.60
Persists in one's actions despite the risk of personal harm or detriment	7.57
Perseveres in spite of great personal risk	7.56
Does not back down from worthy causes	7.54
Fights for what is true and just	7.53
Does things in spite of personal sacrifice	7.52
Does not give up	7.52
Takes responsibility for actions, even when difficult	7.52
Acts bravely in dangerous situations	7.51
Does not back down when scared	7.50
Acts bravely	7.50
Takes risks for a noble end	7.49
Sacrifices one's life for another person's survival	7.48
Does not quit when “the going gets tough”	7.46
Endures hardship if necessary to achieve ideals	7.45
Sacrifices self to save another	7.45
Follows through even if one is scared	7.44
Stands up for what one believes is right	7.44
Remains steadfast in one's beliefs in the face of persecution	7.44
Explores unfamiliar territory despite fear	7.43
Risks their life for a greater good	7.43
Stands up to unjust social practices because of what one thinks is right	7.43
Does what is right even if it is unpopular	7.43
Accomplishes something despite personal fear	7.40
Does not abandon others in dangerous situations	7.40
Acts in the face of adversity	7.40
Endures hardship if necessary to achieve mission	7.39
Risks death to do something one knows is necessary	7.39
Behaves in accordance with one's beliefs, even if the beliefs are unpopular	7.39
Overcomes challenges	7.37
Acts upon one's values even if the values are not the norm	7.36
Defends those who cannot defend themselves	7.36
Says no to what is wrong when it is not popular	7.36
Stands one's ground even if it goes against popular opinion	7.36
Fights through pain and suffering	7.35
Stands up for what one believes no matter what the circumstances	7.33
Takes responsibility	7.33
Risks life to protect others	7.33
Takes responsibility for one's mistakes	7.33
Stands up for the “right thing” against opposition	7.33
Displays perseverance	7.33
Tells the truth even if there may be negative consequences	7.32
Stays and helps when others run	7.32
Confronts danger	7.31
Endures tough situations	7.30
Speaks up for one's beliefs, even under controversy	7.30
Does not surrender under coercive conditions	7.29
Makes the best possible choice to better everyone, not just self	7.28
Perseveres in face of obstacles	7.28
Rises to face daunting challenges	7.27

Results

Based on remarkable similarities between Yale University and USAFA samples ascertained in the previous studies, both hierarchical cluster analysis (HCA) and CMDS used combined samples to confirm previous findings (also based on combined samples) regarding the implicit underpinnings of courage. HCA and CMDS were applied to the same dissimilarity matrix of variances. HCA provides an empirical method for determining the hierarchical structure of prototypical categories or concepts (Rosch, 1978). CMDS allows one to determine the perceived relative representation, based on an appropriate number of dimensions, of the items in a matrix (Hair, Anderson, Tatham, & Black, 1998). We used both analytic techniques to discover underlying structures in our data. HCA and CMDS do not, however, explain why the clusters or dimensions exist.

Hierarchical cluster analysis. Cluster analysis was performed on the variance dissimilarity matrix using complete linkage (furthest neighbor) amalgamation rules. While there is no established objective procedure for determining the optimal number of clusters to retain from HCA, one can look at the coefficients in the agglomeration schedule. The coefficients indicate the similarity/dissimilarity between two clusters (or items) joined at each stage. One looks for a jump in the coefficients. The stage before the sudden change indicates the optimal stopping point for merging clusters (Norušis, 2005). As with factor analysis and CMDS, interpretability of the clusters is another important consideration when deciding on the optimal number of clusters. Following this guidance, we found a three-cluster solution to be appropriate.

Three distinct clusters and their associated behaviors form the cluster solution for courage. The first cluster was labeled *self-sacrifice/risk for others* ($n = 14$ behavioral items), and included behaviors such as “sacrifices self to save another” and “risks one’s life to save others in need.” The second cluster was labeled *non-physical/social-oriented acts for noble ends* ($n = 24$). It included behaviors such as “speaks up for one’s beliefs, even under controversy,” “acts upon one’s values even if the values are not the norm,” and “does what is right even if it is unpopular.” The third cluster was labeled *self-focused perseverance despite fear* ($n = 22$). This cluster included behaviors such as “does not back down when scared,” “accomplishes something despite personal fear,” “displays perseverance,” and “endures hardship if necessary to achieve ideals.”

To determine the relative importance of each cluster to the participants, we averaged the

prototypicality ratings for the behaviors (generated in Study 2) within a cluster. The means for the three clusters were 7.48, 7.49, and 7.43, respectively. A one-way analysis of variance was conducted on the three cluster means. Results indicated no significant difference between the perceived importance of each cluster.

Classical multidimensional scaling. We analysed the scaling of the dissimilarity matrix using CMDS with the ALSCAL program. A non-metric, Euclidean distance model was specified, along with ordinal, matrix conditional, untied data. The appropriate number of dimensions was determined by Kruskal’s Stress Formula 1, which shows the degree of “badness” of fit between the model and scaled data and R^2 , which measures the proportion of matrix data variance accounted for by the model. In addition, Kruskal and Wish (1978) recommended that the number of items minus one should be at least four times as great as dimensionality for ensuring statistical stability of the analysis.

Table IV shows the CMDS scaling for courage. Three dimensions accounted for 94% of the variance in the data, with stress of 0.09. The three-dimensional solution tended to yield bipolar dimensions in which the positive and negative polarities lent themselves to separate interpretations. Such bipolar dimensions are common in this kind of work (Sternberg, 1985). The first dimension, means to achieve noble ends, yielded two interpretations: physical risk/sacrifice for the positive polarity (e.g., risks one’s own life to save others in need; sacrifices self to save another; risks one’s life for a greater good) and non-physical/social-oriented action for the negative polarity (e.g., acts upon one’s values even if the values are not the norm; stands up for what one believes in the face of strong opposition).

The second dimension, self versus other—motivation for action, also yielded two bipolar interpretations: other-focused motivation for the positive polarity (e.g., does the right thing regardless of consequences to self; sacrifices self for a cause; risks one’s life for a greater good) and self-focused motivation for the negative polarity (e.g., explores unfamiliar territory despite fear; follows through even if one is scared; overcomes challenges).

The third dimension, responsibility and enduring action, further yielded two bipolar interpretations: responsibility for actions for the positive polarity (e.g., takes responsibility even when difficult) and endurance/perseverance for the negative polarity (e.g., endures hardship if necessary to achieve ideals; displays perseverance; does not give up). Figure 1 shows the conjoining of the HCA and CMDS techniques.

Table IV. Nonmetric multidimensional scaling solutions for courageous behaviors.

Data label	Scaling solutions	Stimulus coordinates
Dimension 1 (<i>Means to achieve noble ends</i>)		
Positive polarity: <i>Physical risk/sacrifice</i>		
V60	Risks one's own life to save others in need	1.87
V1	Risks one's own safety to save others in need	1.84
V55	Sacrifices one's life for another person's survival	1.82
V56	Sacrifices self to save another	1.80
V8	Sacrifices one's life for another person's safety	1.80
V59	Risks one's life to protect others	1.75
V35	Risks death to do something one knows is necessary	1.55
V51	Sacrifices self for a cause	1.54
V31	Does not abandon others in dangerous situations	1.49
V29	Risks one's life for a greater good	1.37
Negative polarity: <i>Non-physical/social-oriented action</i>		
V36	Behaves in accordance with one's beliefs, even if the beliefs are unpopular	-1.83
V46	Speaks up for one's beliefs, even under controversy	-1.81
V33	Acts upon one's values even if the values are not the norm	-1.81
V6	Stands up for what one believes in the face of strong opposition	-1.74
V39	Stands one's ground even if it goes against popular opinion	-1.74
V10	Stands up for what is right even if the majority disagrees	-1.68
V57	Says no to what is wrong when it is not popular	-1.65
V37	Stands up for what one believes no matter what the circumstances	-1.64
V26	Does what is right even if it is unpopular	-1.61
V25	Stands up to unjust social practices because of what one thinks is right	-1.61
Dimension 2 (<i>Self vs. other—motivation for action</i>)		
Positive polarity: <i>Other-focused motivation</i>		
V50	Makes the best possible choice to better everyone, not just oneself	1.72
V9	Does the right thing regardless of consequences to self	1.63
V41	Defends those who cannot defend themselves	1.36
V43	Tells the truth even if there may be negative consequences	1.36
V29	Risks one's life for a greater good	1.28
V4	Does what is right even if no one will know	1.13
V18	Does things in spite of personal sacrifice	1.11
V51	Sacrifices self for a cause	1.09
V8	Sacrifices one's life for another person's safety	1.04
V60	Risks one's own life to save others in need	1.01
Negative polarity: <i>Self-focused motivation</i>		
V24	Explores unfamiliar territory despite fear	-1.69
V38	Overcomes challenges	-1.61
V17	Does not give up	-1.56
V21	Does not quit when "the going gets tough"	-1.48
V28	Follows through even if one is scared	-1.48
V20	Does not back down when scared	-1.44
V30	Accomplishes something despite personal fear	-1.43
V53	Rises to face daunting challenges	-1.42
V44	Displays perseverance	-1.37
V48	Endures tough situations	-1.36
Dimension 3 (<i>Responsibility and enduring action</i>)		
Positive polarity: <i>Responsibility for actions</i>		
V14	Takes responsibility for actions, even when difficult	1.47
V40	Takes responsibility for one's mistakes	1.45
V45	Takes responsibility	1.28
Negative polarity: <i>Endurance/perseverance</i>		
V22	Endures hardship if necessary to achieve ideals	-1.44
V44	Displays perseverance	-1.35
V21	Does not quit when "the going gets tough"	-1.23
V13	Does not back down from worthy causes	-1.18
V48	Endures tough situations	-1.01
V17	Does not give up	-1.00

Note: Proposed labels italicized; stress (Kruskal's stress formula 1) = 0.09 $R^2 = 0.94$.

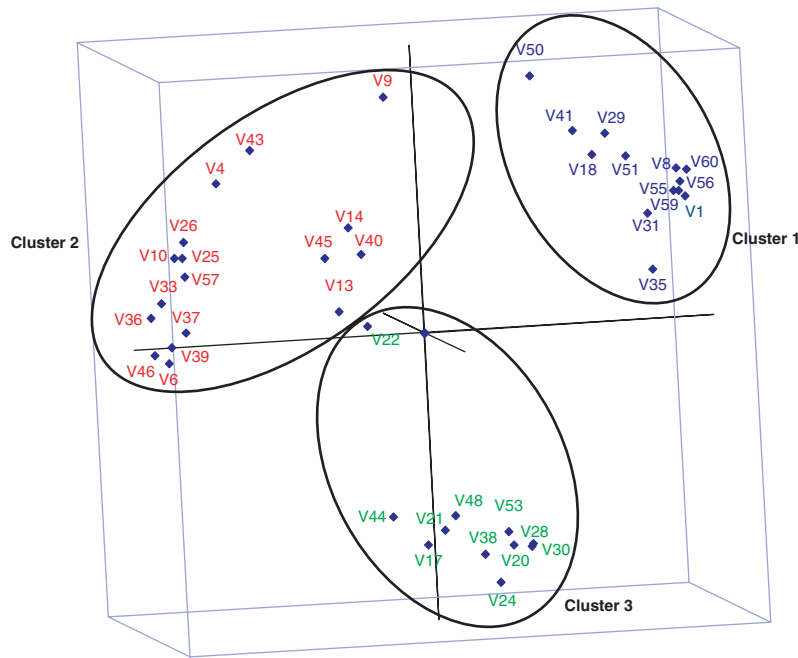


Figure 1. Conjoining hierarchical cluster analysis and multidimensional scaling dimensions. Three distinct clusters are represented. Data labels correspond to behaviors listed in Table IV.

Discussion

We determined several notable aspects of this study. First, the proportion of variance in the CMDS three-dimensional solution accounted for in the data was 0.94, with stress of 0.09. These indices of fit indicated the scaling was very reliable, as the model extracted most of the variance from the proximity data with a small number of dimensions. It is therefore reasonable to conclude that the dimensions obtained reflect people's notions of the underlying structure of courage rather than spurious or random fluctuations of the data.

Second, as one should expect, the three-cluster HCA solution and the three-dimensional CMDS solution provided complementary evidence for, and clear, interpretable representations of, people's implicit structures of courage. Figure 1 clearly indicates a complementary relationship between the two techniques. Further, as evidenced by HCA, people found each cluster to be of similar importance in representing the concept of courage. These analytical techniques uncovered notable behaviors illustrative of the concept of courage, such as risking or sacrificing self, persisting despite fear, and acting for a noble/worthy purpose.

A third notable aspect of Study 3 results was their relationship to this article's Study 2 results. The clusters and dimensions complemented the factors obtained in Study 2 with remarkable consistency, which used a different methodology (factor analysis of the distinctively characteristic ratings), a different

set of participants, and a superordinate set of 131 behaviors. The three HCA clusters (self-focused perseverance despite fear; non-physical/social-oriented acts for noble ends; and self-sacrifice/risk for others) corresponded quite consistently with the three Study 2 factors (self-focused response to affect and external circumstances; non-physical/social-oriented acts—internal noble motivation; and selfless sacrifice/risk—external noble motivation). The concern then, that participants merely sorted behaviors based on the presence or absence of key words (e.g., fear, risk) appears unfounded in light of this remarkable consistency between studies. Overall, the implicit and theoretical underpinning of courage as obtained in Studies 2 and 3 are very consistent.

Fourth, and arguably most important, the dimensions and clusters appear clearly to represent people's notions about the nature of courage. The purpose of the present study's analysis, as Sternberg (1985) would say, is "not to go beyond what we already know, but rather to bring what we know into the open for scientific inspection" (pp. 617, 618). When we characterize someone's behavior as courageous or evaluate someone as possessing "courage," we must be prepared to support what we mean by the use of these terms.

Study 4

In this study, we sought to ascertain the extent to which people actually used behaviors associated with courage to evaluate other people's courage,

particularly when they are presented with written descriptions.

Method

Participants. In this study, 169 undergraduate and graduate students from the USAFA and Yale University participated. The sample included 92 men and 72 women; ages ranged from 18 to 36 years of age ($M = 21.9$, $SD = 2.9$). Participants received either course credit or US\$10 compensation for their participation.

Materials. Participants received an informed-consent document and completed a brief online questionnaire eliciting demographic information. Participants then completed the vignette-rating task. All participants received the same 33 vignettes. The vignettes described scenarios of varied situations and individual behaviors and differed from one another only in that different pairing of names with descriptions of people were used for different participants (i.e., a given vignette was paired half of the time with a male name and the other half with a female name). Vignettes were developed, in part, from past real-world events, while other scenarios were entirely fictitious. Each vignette was developed with varying levels of four dimensions (i.e., intentionality/deliberation; presence of personal fear; noble/good act; and known substantial personal risk). These thematic dimensions were revealed in our previous studies.¹ These dimensions were manipulated along a continuous scale rather than discrete (present/absent) variations. Not all combinations of varying levels of the dimensions could be represented in the vignettes; however, efforts were made to avoid correlated pairing levels of dimensions. The following two examples are typical vignettes used in this study:

- (1) Mark (Julie) was surfing off the coast of the Florida panhandle when he witnesses a 14-year-old girl being attacked by a tiger shark. The young victim was in severe agony as she screamed for help. Although in stunned disbelief and visibly shaken by what he was witnessing, Mark immediately began paddling as hard and as quickly as he could to reach the girl. Mark began to hit, beat, and kick the tiger shark to ward off further attacks. He was able to beat back the 7-foot attacker. He then pulled the girl onto the surfboard and paddled to shore.
- (2) Susan (Roger) had been climbing mountains for a long time. Now she had the opportunity to climb a 14er (mountain over 14,000 feet in elevation). Four experienced climbers had died during the previous year attempting this

same feat. This was the last mountain on Susan's list. Nothing was going to stop her from making this climb. The climb was fraught with unstable conditions, both from a geological and meteorological standpoint, but Susan dauntlessly succeeded in reaching the summit.

In real life, evaluations of an individual's courageous behavior would be largely subjective and require the observer to make certain inferences (e.g., level of fear of the protagonist). Therefore, we maintained the ecological validity of the vignettes by avoiding overly formulaic designs. Vignettes were descriptive of "physical," "moral," and "social" situations; however, this study did not explicitly compare these different "brands" or types of courageous behavior.

Procedure. Participants spent approximately 30 minutes completing the computerized, web-based, vignette-rating task. In an effort to assure independence of dimension and courage ratings, one half of the participants were instructed to read the vignettes and then to evaluate the presence of four dimensions. The other half of the participants were instructed to read the vignettes and then to evaluate the vignette protagonist's level of courage. Participants were randomly assigned to each condition. All ratings were made on a 1 (low) to 9 (high) scale. All responses were electronically recorded.

Results

Initially, we conducted $2 \times 2 \times 2$ factorial ANOVAs separately for the four dimensions and courage ratings. The independent variables included academic institution (USAFA vs. Yale), participant gender (male vs. female), and vignette name (male vs. female). All main effects and interactions failed to reach significance.

Next, we collapsed the data across participants (using vignettes as the unit of analysis) for reliability and regression analyses. For each vignette the rating of courage was treated as the dependent variable, and the ratings of the four dimensions were the independent variables. Table V presents the results of this study. The first panel displays the basic statistics of the variable ratings along with the Spearman-Brown split-half subject and item reliabilities of the ratings. All ratings were extremely reliable in terms of subject reliability and respectably reliable in terms of item reliability.

The intercorrelations among the variables, displayed in the second panel, show strong, positive associations between the four dimensions and courage. In particular, we noticed a strong

Table V. Results for “vignette-rating” study.

Variable	M	SD	Subject reliability ^a	Item reliability ^a	
Basic statistics					
Intentionality	6.73	1.55	0.98		0.78
Fear	5.26	1.71	0.98		0.77
Noble purpose	6.13	2.35	0.99		0.70
Risk	5.96	1.85	0.98		0.75
Courage	5.28	1.83	0.99		0.81
	Intentionality	Fear	Noble purpose	Risk	Courage
Intercorrelations of ratings					
Intentionality	1.00	0.40*	0.09	0.55**	0.44**
Fear		1.00	0.03	0.68**	0.31
Noble purpose			1.00	0.16	0.80**
Risk				10.00	0.57**
Courage					1.00
Variables	B	β	sr^2		
Standard multiple regression coefficients					
Intentionality	0.22	0.19*	0.02	Intercept = −1.65	
Fear	−0.04	−0.04			
Noble purpose	0.56	0.72**	0.50		
Risk	0.38	0.38**	0.06		
Courage (DV)					

$R = 0.93^{***}$ $R^2 = 0.86^b$ Adj $R^2 = 0.84$

^aReliability was calculated using Spearman–Brown split-half formula on randomly assigned split halves. ^bUnique variance = 0.59; shared variance = 0.27. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

association, $r = 0.68$, $p < 0.01$, between the presence of personal fear (Fear) and known substantial personal risk (Risk). The correlations between male and female vignette protagonist ratings for each variable were all in the mid- to high-90s. This set of results indicated that the pattern of ratings was very similar, regardless of the gender of the vignette protagonist. Furthermore, the mean ratings of courage were equal to the hundredth place for male and female names ($M = 5.28$). The ratings of the four dimensions were also similar for male and female names, respectively: intentionality ($M = 6.75$, 6.67), Fear ($M = 5.22$, 5.42), Noble Purpose ($M = 6.12$, 6.10), and Risk ($M = 5.88$, 6.06). Therefore, the analysis shows that the levels of ratings as well as rating patterns were the same across the genders of the vignette protagonists.

Finally, multiple regression was used to predict the ratings of courage from the rated presence of the four dimensions (Intentionality, Fear, Noble Purpose, and Risk). Regression analysis results appear in the third panel. The regression analysis indicated a significant $R = 0.93$, $F(4, 28) = 43.86$, $p < 0.001$. The total variance in courage accounted for by knowing the ratings of the four dimensions was 86%. The precision index, or standard error of estimate for this model, was 0.73. This regression model did exceptionally well in predicting levels of courage of the protagonists depicted in the vignettes. Three of the four independent variables contributed significantly to prediction of courage.

Intentionality ($sr^2 = 0.02$), Noble Purpose ($sr^2 = 0.50$), and Risk ($sr^2 = 0.06$) accounted for a unique variance of 0.59. The four dimensions combined contributed another 0.27 in shared variability. Although the correlation between Fear and courage was marginally significant, $r = 0.31$, $p = 0.075$, Fear did not contribute significantly to the regression. It is possible that the relationship between Fear and courage is redundant to, or mediated by, the relationship between Risk and courage.

Discussion

One might have expected dissimilar responses and evaluations of courage from such disparate samples; a more male-oriented, conservative/traditional sample (USAFA) and a more female-oriented, liberal sample (Yale University). Statistical differences were not demonstrated by our analyses eschewing the notion that these samples may perpetuate a certain type of courage. Regardless of gender or academic institution, our participants similarly used their implicit theories of courage in evaluating the courage of others. Their evaluations were based on relatively brief scenarios and relatively little information, yet these judgments could be predicted at high and accurate levels on the basis of their implicit theories of courage. Furthermore, participants appeared to use similar implicit theories in evaluating both male and female protagonists. This finding alone is interesting as it is in contrast with traditional

images and explicit studies of men as the sole exemplars of courageous behavior (Aristotle, trans. 1987; McMillan & Rachman, 1987; Miller, 2000; Plato, trans. 1987; Rachman, 1990).

The regression model indicated that people evaluate the levels of intentionality of action, nobility of purpose, and known substantial personal risk when assessing others' behaviors as courageous. However, the presence of personal fear did not contribute to the regression model, as expected. This finding is conceptually contrary to previous implicit-theory studies (Evans & White, 1981; O'Byrne et al., 2000; Philips, 2004; Woodard, 2004). These results are also contrary to our own previous findings. Studies 1 and 2, for example, provided empirical support for fear as a core component in people's implicit theories of courage. Our factor analysis, HCA, and CMDS all revealed that "acting despite fear" was integral to people's conceptions of the meaning of courage.

At least three possibilities could account for this inconsistency. First, although people report fear or acting despite fear as integral to their conceptualizations of courage, this component may not play a necessary or core role in evaluating others' levels of courage. Walton (1986) dismisses the emotion of fear as a "necessary" condition of courage, and Shelp (1984) also indicates by his equivocation, "perhaps fear," that fear may not be necessary in determining or defining what courage is.

Second, possibly we did not correctly anticipate how fear would be perceived in the vignettes. Fear and Risk were highly correlated, $r = 0.68$, $p < 0.01$ and perhaps redundant. Participants may have been unable to separate the affective component (personal fear) from the conditions or circumstances (known substantial risk). Future studies should make every effort to disentangle fear and risk to ensure participants' accurate understanding of the dimensions.

Third, evaluating the level of fear in others may be too subjective and, by default, one's own fear of the situation becomes the contributing factor. In other words, it may not be the level of fear of the protagonist that is predictive of courage, rather the level of fear of the observer projected on to the vignette. Evans and White (1981) came to a similar conclusion. Participants in their study attributed bravery (courage) to the target actor if the actor was doing something that would have frightened the participant. Another notable aspect of this study was the large contribution of noble/good purpose to the regression model. The regression model indicated that, in order for our vignette protagonists to be evaluated as highly courageous, the motivation for acting must be present and identifiable as noble (worthy, good) intentions and ends. Some have

argued this ethical dimension is what separates the concept of courage from other concepts and behaviors, such as bravery and intrepidity (Walton, 1986), recklessness (Woodard, 2004), and thrill seeking (Gould, 2005).

In sum, people do use their implicit theories, at least in part, to make judgments and evaluations of others' courage, and these judgments can be predicted from such implicit theories, as was done in this study. This study also revealed that the role of an affective component (e.g., fear) in courage needs to be further evaluated in terms of its disentanglement with risk and in terms of actor versus observer levels.

General discussion

What do people think courage is? Those investigating the psychology of courage have yet to reach consensus. Despite the vast number and wide-ranging definitions of courage, we endeavored to unveil a common core to this psychological construct. This line of inquiry sought to discover the nature and use of people's implicit theories of courage.

However, does the implicit approach really advance the field of courage studies, when explicit/scholarly definition of courage abound that demand further study? The answer to this question is an unequivocal yes. The implicit-explicit dichotomy debate has previously emerged in several domains such as intelligence (Sternberg et al., 1981) and wisdom (Sternberg, 1985). Good operational definitions are important. Implicit- and explicit-theory research exemplifies two components of a more complete operational definition and approaches the objective of a universal definition of courage from different angles. In reference to implicit theories, Bluck and Glück (2005) state that "the notion that an entire culture would carry around an organized construct of wisdom [courage] that is wrong, or incorrect, is a slightly bizarre one" (pp. 90, 91). It should also be recognized that explicit definitions of courage often differ among researchers, and that these definitions are, in part, the researcher's implicit theories of the construct. Though we have not reached consensus, there remains substantial convergence between implicit and explicit research on similar sets of dimensions as central to descriptions and definitions of courage.

People are able to articulate well-developed, implicit theories of courage. Although the pioneering work of O'Byrne et al. (2000) indicates myriad views and descriptions of courage, they recognize emerging implicit-theory notions such as taking risks, attitudes, facing challenges, and defending beliefs. Greatly expanding on their foundational work, our three

initial studies indeed confirm the presence of four necessary components of people's notions of courage; that is, intentionality/deliberation, personal fear, noble/good act, and known personal risk.

In Study 4 we discovered that people actually use behaviors associated with courage when evaluating other people's courage, particularly when they were presented with written descriptions. In fact, we predicted courage very accurately by knowing people's ratings on three of the four dimensions (intentionality/deliberation, noble/good act, and known substantial personal risk). Protagonist levels of personal fear did not significantly contribute to the model.

It is interesting to note that there still remains a long-standing debate over the role of fear in courage. Two key historical figures in this debate were Aristotle and Plato. Aristotle (trans. 1987) regarded courage as the disposition to act appropriately in situations that involve "feeling" fear. On the other side of the debate, Plato (trans. 1956), from *Protagoras*, described courage as "the knowledge of what is and is not to be feared." Plato described a rational understanding of fear rather than an emotional/affective feeling of fear as a component of courage. The fact that people have held these disparate notions of courage for centuries could help explain the inconsistencies between Studies 1–3 and Study 4. Recognizing this complexity could help inform future thinking and study of the role of fear in courage.

Do we now know what courage is? Perhaps we are closer than we had previously thought. As we review the literature which covers a variety of empirical approaches to people's conceptions of courage, we find that at least four aspects of courage emerge (namely, risk, fear, intentionality/deliberation, and noble good). The results of the present research seem to converge with these findings. This also suggests that certain core behaviors of courage emerge regardless of the empirical methodology. The results of the present study also seem to converge on the more recent explicit definition of courage described by Peterson (2006), Peterson and Seligman (2004) and Snyder and Lopez (2007) as "the exercise of will to accomplish goals in the face of opposition, either external or internal." Parsing their definition indicates similar components that are central to the definition of courage, such as intentionality/deliberation (exercise of will), noble good (accomplish goals), and known substantial risk (in the face of opposition). To reiterate, there is at least tacit agreement that courage is a complex, multi-dimensional construct.

Using our studies as a basis to know what courage is, and the present studies' convergence with the existing literature, we might best describe or conceptualize courage as: (a) a willful, intentional act, (b) executed after mindful deliberation, (c) involving objective substantial risk to the actor, (d) primarily motivated to bring about a noble good or worthy end, (e) despite, *perhaps*, the presence of the emotion of fear. In general this description of courage is comprised of situational (external circumstances), cognitive, volitional, affective, and motivational components. Perhaps there are additional components, but those components may be peripheral to the meaning of courage. However, if a core dimension of courage is absent (understanding that *fear* may not be a necessary core component), then one could not appropriately claim a complete definition or description of courage. Additional studies are required to address this particular issue.

As with any study, this one comes with its limitations. This study was not developmental in its approach to implicit theories of courage. Szagun (1992) noted differing notions of courage in a cross-sectional study of children in three age groups: 5–6, 8–9, and 11–12 years old. Older children appear to understand courage in increasingly psychological and nonphysical terms. Longitudinal studies, however, are required to show whether or not these implicit theories of courage change over the life span. Szagun's study, as with this one, was fixed at a moment in time. We cannot provide evidence to support or refute the notion that implicit theories of courage change over time. These studies can only be interpreted as accurately reflecting the implicit theories of the participants at the time it was conducted. Further, our study was limited to one particular culture, rather than being cross-cultural. While our results may generalize broadly within our culture, it is reasonable to assume, based on the vast number of definitions of courage alone, that conceptions of courage may differ from culture to culture. Another possible limitation of this study was in terms of the skewed sample of graduate versus undergraduate participants (Yale sample), and in terms of gender (USAFA sample). Researchers may also view our samples as too "rarified." Future studies could employ the use of stratified community samples with ample participants to further gauge the "every-person" view of courage.

Future directions

Given the limitations and results of the present study, several paths for future research should

be considered. First, it is important, whether conducting explicit or implicit research, that appropriate samples are used. The results of the present study could be validated by first using stratified community samples, then moving to cross-cultural samples to garner broader views of the construct of courage. Although the present study is concerned with finding a more global conceptualization of courage, specialized samples could target particular and specific nuances.

In terms of methodologies, ecologically valid studies might be conducted by developing vignettes of courage using the methodology proposed by McFall and Lillesand (1971) for their Conflict Resolution Inventory. This procedural variation could accomplish several things: (a) increase the relevance to the participant's daily lives; (b) increase the ecological validity with respect to everyday acts of courage experienced or witnessed by the sample participants; (c) increase the potential for intervention applications of the findings; and (d) improve the assessment of realistic courageous behaviors. In addition, Frisch and Higgins (1986) describe efforts to behaviorally and unobtrusively assess assertiveness that could be applied to the assessment of courage. As Deutsch (1961) mentioned, it is difficult, however, to assess the multi-dimensional complexities of courage in a laboratory setting.

Given the present study's inconsistent results with regards to the role of fear (however consistent with the long-standing major debate) researchers should continue to investigate fear, and emotion in general, as it relates to courage. Investigations should focus on courage and (a) not feeling fear at all, (b) feeling fear but overcoming it in order to act, and (c) acting with but despite of fear.

Finally, given the implicit-explicit dichotomy debate, Rate is conducting an analysis that seeks to integrate the two views and the myriad conceptualizations of courage into a broad, umbrella conceptualization. This conceptualization will initially be investigated in future vignette-based studies. The integration of scholarly implicit- and explicit-theory definitions should provide a solid springboard to then singularly investigate the components of courage and peripheral influences that have encouraged branding several types of courage.

Conclusion

What do people *think* courage is? This study has provided partial answers to this question. Those seeking to behave in a courageous way may find it useful to know what is considered courageous, but also, what is not. One problem this research will not

resolve is whether any particular act is or is not courageous because there will always be disagreement as to the "necessary and sufficient" threshold of its components, such as what is a noble or worthy purpose. Such disagreements, regrettably, are the basis of much of the conflict in the world; past, present, and, in all likelihood, future.

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Notes

1. The addition of a volitional, cognitive dimension, intentionality/deliberation, was included as it had a strong presence in the Study 1 response-generation task. It was determined that we erroneously dropped this component as a modifier (e.g., willingly) early on in this sequence of studies. We later determined that this dimension should be included as a potential "necessary" dimension of courage. For these same reasons, we added the modifier "known" to personal risk explicitly qualifying a cognitive connection of "awareness" to risk.

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